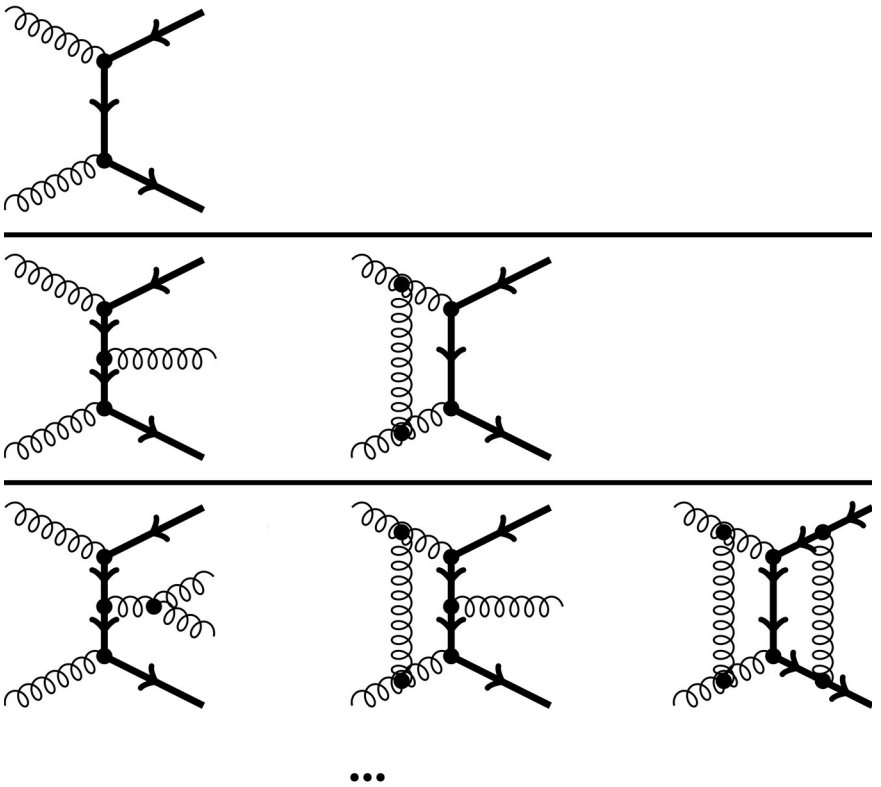
Quantum Chromodynamics

René Poncelet

Lecture 3:
systematics of higher-order computations &
phenomenology

THE HENRYK NIEWODNICZAŃSKI
INSTITUTE OF NUCLEAR PHYSICS
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European School of High Energy Physics
19th June - 3rd Jul 2026 - Ostroda



Lecture outline

- Motivation
- The QCD Lagrangian
  + running of the strong coupling
- Infrared divergencies
- Parton-distribution functions
- Anatomy of a higher-order QCD computation
  - Virtual/Loop corrections
  - Real emission corrections
- **Systematics of higher-order computations**
- **Phenomenology**

Systematic higher order computations

Systematics of loop computation

In principle we have all the building blocks to compute any loop diagram

but in practise algebra challenging → own line of research:

- number of Feynman diagrams grows fast with the loop-order and number of legs
- more particles lead to **many** kinematic scales … even more algebra
- how to evaluate appearing Feynman integrals?

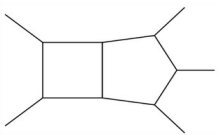
Two important techniques
(you often will hear about in TH talks)

→ **Integration-by-parts Identities**

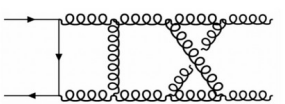
→ **Master integral differential equations**

State-of-the-art box

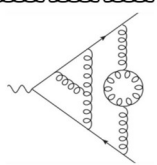
Two-loop 5-point
[Abreu, Agarwal, Badger, Buccioni, Chawdhry, Chicherin,Czakon, de Laurenties, Febres-Cordero, Gambuti, Gehrmann, Henn, Lo Presti, Manteuffel, Ma, Mitov, Page, Peraro, Poncelet, Schabiner Sotnikov, Tancredi, Zhang,...]



Three-loop 4-point
[Bargiela, Dobadilla, Canko, Caola, Jakubcik, Gambuti, Gehrmann Lim, Mella, Mistleberger, Wasser, Manteuffel, Syrrakow, Smirnov Trancredi, ...]

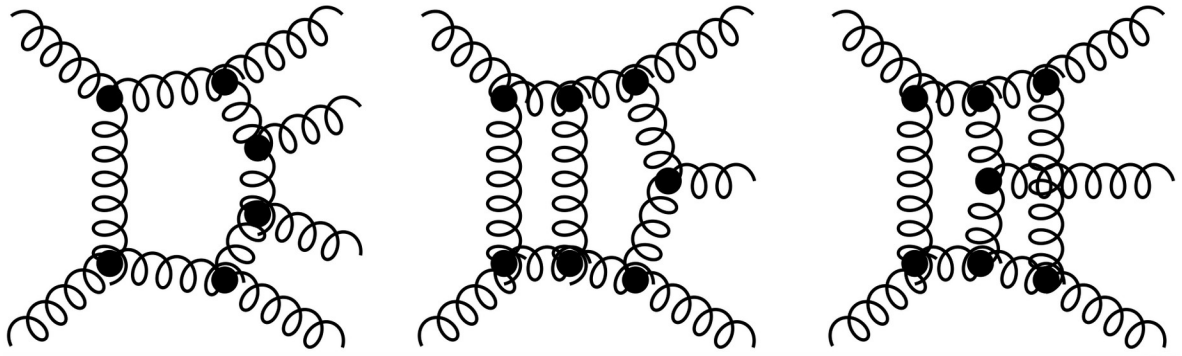


Four-loop 3-point
[Henn, Lee, Manteuffel, Schabinger, Smirnov, Steinhauser,...]



Approach to multi-loop amplitudes

Feynman diagrams



→ many graphs contribute to a given process, O(10^3 – 10^4) typical at the moment

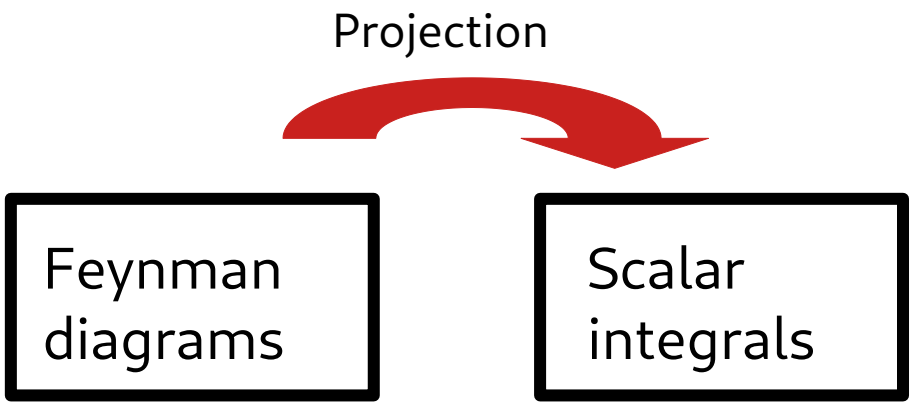
→ however fully automated techniques available: **QGraph**, **DiaGen**, ...

→ direct evaluation not possible:

→ (tensor-)loop integrals need regularisation (IR & UV) $\rightarrow d = 4 - 2\epsilon$

→ dimensional regularisation obscures spin& Lorentz structure

Color & Spin projection



color & spin decomposition:

color structure

$$\mathcal{M} = \sum_{c,s} \mathcal{M}_{c,s} \mathbf{C}_c \mathbf{T}_c$$

spin structure

scalar partial amplitudes

$$\mathcal{M}_{s,c} = \sum_j c_j(\{p\}, \epsilon) \mathcal{I}_j(\{p\}, \epsilon)$$

rational functions
(kinematics and ϵ)

(scalar) Feynman Integrals

Integration-By-Parts reduction

Rational functions of the kinematic invariants and ε

$$\sum_{\text{color,spin}} \left(\mathcal{M}^{(0)}\right)^{\dagger} \mathcal{M}^{(L)} = \sum_{i} c_i(\{p\},\varepsilon)\, \mathcal{I}_i(\{p\},\varepsilon)$$

Scalar (or Feynman) integrals:

Number of loops

Number of propagators

$$\mathcal{I}_i(\{p\},\epsilon) \equiv \mathcal{I}(\vec{n_i},\{p\},\epsilon) = \int \prod_{l}^{L} \frac{\mathrm{d}^d k_l}{(2\pi)^d} \prod_{m=1}^{N} D_m^{-n_{i,m}}(\{p\},\{k\})$$

D_m : squared propagators, for example $D_1 = (p_1 - k_1)^2$, $D_2 = (k_1 - k_2)^2$, ...

Integration-By-Parts Identities and reduction

Bonus feature of dimensional regularization:

$k_i \to k_i + l$ does not change the integral since the boundaries are at infinity

$$\longrightarrow \quad 0 = \int \prod_l^L \frac{\mathrm{d}^d k_l}{(2\pi)^d} \frac{\partial}{\partial k_n^\mu} l_\mu \prod_{m=1}^N D_m^{-n_{i,m}}(\{p\},\{k\}) \qquad l \in \{p\} \cap \{k\}$$

This leads to large matrix (all coefficients are functions of the invariants and ε)
→ can be solved for small number of so-called master integrals
→ Gaussian elimination → Laporta algorithm

Non-trivial in practise, many refinements in state-of-the-art computations
→ huge computers…

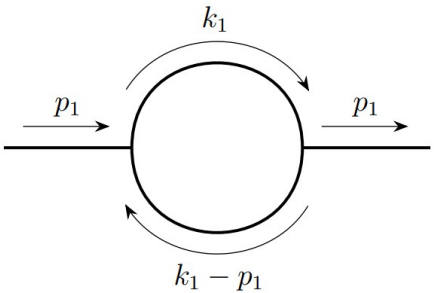
$$\longrightarrow \quad \sum_{\text{color,spin}} \left(\mathcal{M}^{(0)}\right)^\dagger \mathcal{M}^{(L)} = \sum_i c_i'(\{p\},\varepsilon)\ \mathrm{MI}_i(\{p\},\varepsilon)$$

The number of master integrals is typically **much smaller** ($10^6 \to 10^2$)

IBP reduction example

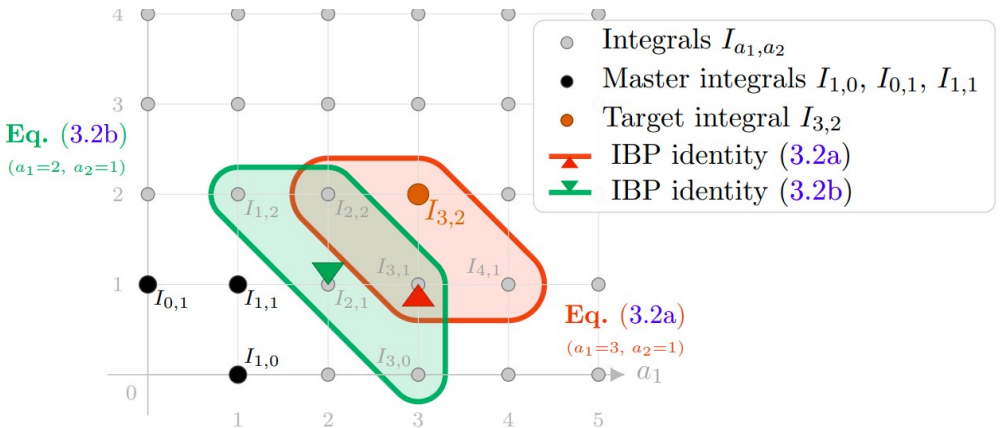
[taken from 2606.10698]

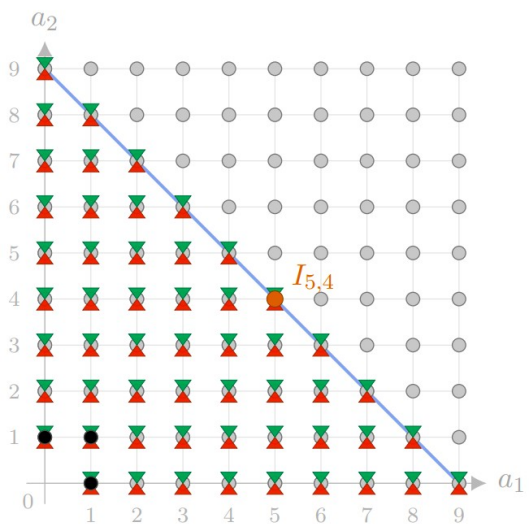
$$I_{a_1,a_2} = \int \frac{d^D k_1}{(-k_1^2 + m^2)^{a_1} (-(k_1 - p_1)^2 + m^2)^{a_2}}$$



$$0 = (D - 2a_1 - a_2) I_{a_1,a_2} + 2a_1 m^2 I_{a_1+1,a_2} - a_2 I_{a_1-1,a_2+1} + a_2 (2m^2 - p_1^2) I_{a_1,a_2+1} \,,$$

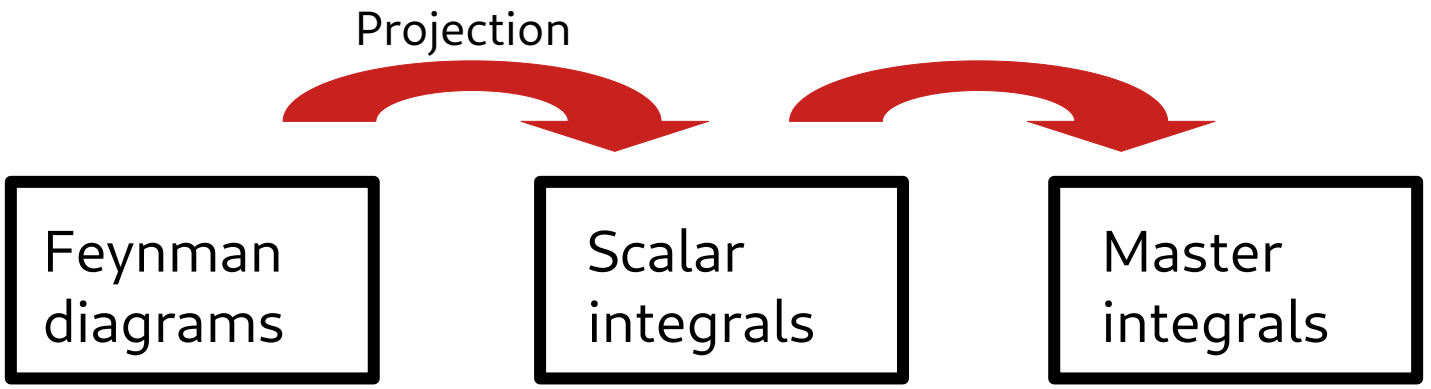
$$0 = (a_2 - a_1) I_{a_1,a_2} + a_1 I_{a_1+1,a_2-1} - a_2 I_{a_1-1,a_2+1} + a_1 p_1^2 I_{a_1+1,a_2} - a_2 p_1^2 I_{a_1,a_2+1}$$





Modern solvers: **Reduze** [von Manteuffel, Studerus], **Fire** [Smirnov, Zeng], **Kira** [Lange, Usovitsch, Wu], **LiteRed** [Lee], **FiniteFlow** [Peraro]
Simplified systems: **NeatIBP** [Z. Wu, J. Boehm, R. Ma, H. Xu, Y. Zhang], **Blade** [Guan,Liu,Ma,Wu], **CALICO** [Bertolini, Fontana, Peraro]

Integration by Part Identities



$$\mathrm{MI}_i(\{p\},\epsilon) \equiv \mathrm{MI}(\vec{n_i},\{p\},\epsilon) = \int \prod_l^L \frac{\mathrm{d}^d k_l}{(2\pi)^d} \prod_{m=1}^N D_m^{-n_{i,m}}(\{p\},\{k\})$$

Typically with smaller $\max\{n_i\}$ then generic integrals → easier integrals

How to deal with the master integrals?
→ **Feynman parameters** and related techniques
→ **Method by regions**: decompose the integral into its IR singularity structure
However, direct analytical integration is **often not possible**:
→ numerics
→ **differential equations**

Differential equations for master integrals

$$\mathrm{MI}_i(\{p\},\epsilon) \equiv \mathrm{MI}(\vec{n_i},\{p\},\epsilon) = \int \prod_l^L \frac{\mathrm{d}^d k_l}{(2\pi)^d} \prod_{m=1}^N D_m^{-n_{i,m}}(\{p\},\{k\})$$

Consider the derivative with respect to external kinematic invariants:

$$\frac{\mathrm{dMI_i}}{\mathrm{d}s_{ij}} = \sum_l \frac{\partial p_l}{\partial s_{ij}} \frac{\partial \mathrm{MI_i}}{\partial p_l} = \sum_k c_k \mathrm{MI_k}$$

$$D_1^{-1} = (p_1 - k_1)^{-2} \rightarrow \frac{\partial D_1^{-1}}{\partial p_1} \sim D_1^{-2}$$

Recover Feynman integrals
→ use IBPs again

System of differential equations: $\mathrm{dMI}_i = \mathrm{d}A(\{p\},\epsilon)_{ij}\mathrm{MI}_j$

→ needs boundary conditions: kinematic limits, special points, numerics,….

→ often solved as expansion in ε

→ analytic solutions → special functions (Polylogarithms, Hypergeometric functions, …)

→ solve numerically as evolution from the boundary

Differential equations for Master Integrals

Projection

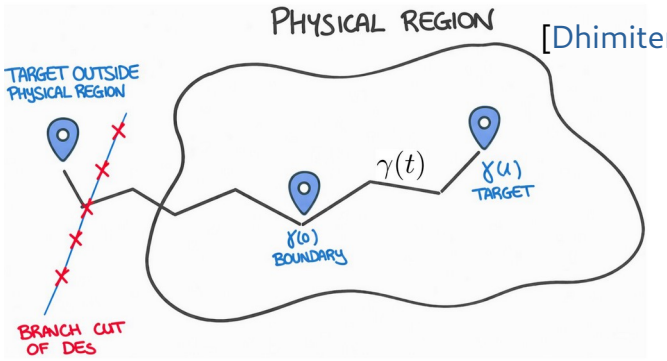
Integration-by-parts

Feynman diagrams → Scalar integrals → Master integrals → 'Analytic' functions

Master integrals fulfil DEQs [Kotikov'91, Gehrmann, Remiddi'00]

$$\partial_x \mathbf{M}_i = \sum_j A_{ij}(\{p\}, \epsilon)\mathbf{M}_j$$

- requires boundary conditions
→ 'analytic' solution if simple otherwise numerics

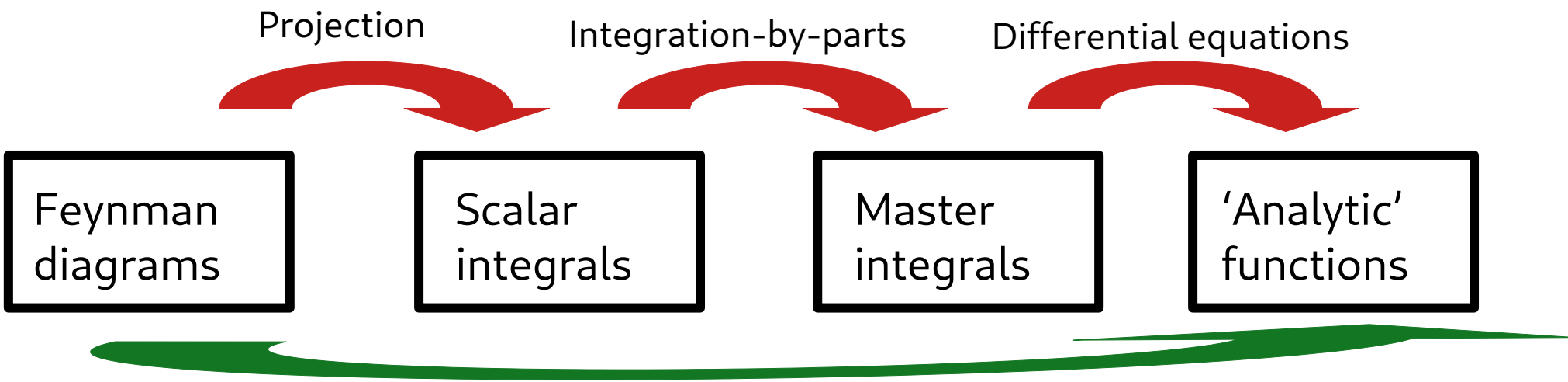


[Dhimiter Canko]

Systematic approach: canonical form [Henn, 13]
→ solutions are iterated integrals

Alternative direct numerical evaluation, e.g. sector decomposition [T. Binoth, G. Heinrich] with **PySecDec** [G. Heinrich, S. P. Jones, M. Kerner, V. Magerya, A. Olsson, J. Schlenk, et al.]

Approach to multi-loop amplitudes



Using **finite field reconstruction** to avoid expression swell:
→ numerical evaluations over finite fields to evaluate at rational points
→ rational interpolation over many points to recover exact analytic expression

[Kant'14, von Manteuffel, Schabinger'14, Peraro'16]
tools: **FiniteFlow**, **FireFly+Kira**, **Fire**, **Caravel**

Automation at one-loop

Only part of loop amplitudes is need in practise

→ needs understanding of the basis of all functions contributing to ε^0 : box, triangles, bubbles

→ 'projection' directly on this basis (up to rational terms)

→ **numerical implementation**: One-Loop-Providers (OLP)

- → OpenLoops
- → Recola
- → MadLoop
- → …

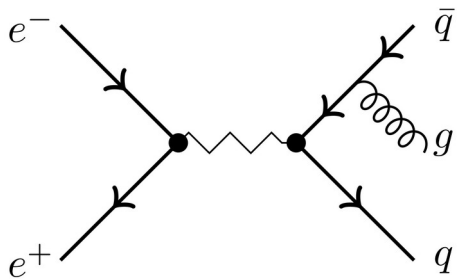
→ **automation at two-loops**?

- → one of the toughest problems of the higher order theory community at the moment

Systematics of real emissions

How to deal with the real radiation contributions?

- phase space constraints make computations more complicated
- more particles imply more soft and collinear limits
- 'observables' might depend on the kinematics
  → how to reconcile this with the phase space integration?



→ eventually only a numerical approach can be practical!
→ but numerics in $d = 4 - 2\varepsilon$ is tricky...

But **how to get rid of the IR divergences**?

Two types of limits: soft and collinear

factorization comes to the rescue...

Systematics of real emissions: soft limits

Factorization of matrix elements in the **soft limit** (only for gluons)

$$\left|\mathcal{M}^{(0)}_{g,a_1,\ldots}(q,p_1,\ldots)\right|^2 \qquad q \to \lambda q \text{ with } \lambda \to 0$$

$$\simeq -4\pi\alpha_s \sum_{ij} \mathcal{S}_{ij}(q) \left\langle \mathcal{M}^{(0)}_{a_1,\ldots}(p_1,\ldots) \middle| \mathbf{T}_i \cdot \mathbf{T}_j \middle| \mathcal{M}^{(0)}_{a_1,\ldots}(p_1,\ldots) \right\rangle \qquad \mathcal{S}_{ij}(q) = \frac{p_i \cdot p_j}{(p_i \cdot q)\,(p_j \cdot q)}$$

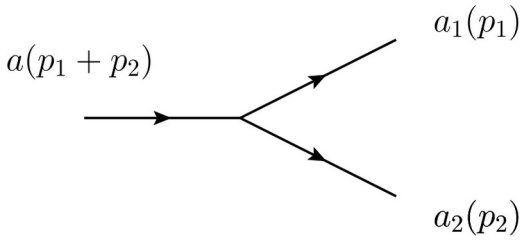
sum over colour-correlators: emissions from colour dipoles

In our e+e- example: $-4\pi\alpha_s \mathcal{S}_{12}(p_3) C_F |\mathcal{M}_2|^2$ for $p_3 \to 0$ $\qquad \mathcal{S}_{12}(p_3) = \frac{1}{2}\frac{s}{s_{13}s_{23}} \sim \frac{1}{\lambda^2}$

[Keep in mind a factor of λ from the phase space measure]

Systematics of real emissions: collinear limits

Collinear limits in the Sudakov parametrization:



$$p_1^\mu = z p^\mu + k_\perp^\mu - \frac{k_\perp^2}{z}\frac{n^\mu}{2p\cdot n}, \qquad p_2^\mu = (1-z)\,p^\mu - k_\perp^\mu - \frac{k_\perp^2}{1-z}\frac{n^\mu}{2p\cdot n},$$

$$s_{12} = 2p_1\cdot p_2 = -\frac{k_\perp^2}{z(1-z)}, \qquad p^2 = n^2 = p\cdot k_\perp = n\cdot k_\perp = 0,$$

$$k_\perp^\mu \to 0.$$

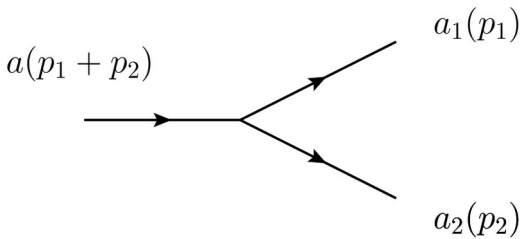
$$\left|\mathcal{M}^{(0)}_{a_1,a_2,\ldots}(p_1,p_2,\ldots)\right|^2 \simeq 4\pi\alpha_s\,\frac{2}{s_{12}}\left\langle \mathcal{M}^{(0)}_{a,\ldots}(p,\ldots)\right|\hat{\mathbf{P}}^{(0)}_{a_1a_2}(z,k_\perp;\epsilon)\left|\mathcal{M}^{(0)}_{a,\ldots}(p,\ldots)\right\rangle$$

diagonal in colour space but gluon – splitting kernels induce spin-correlations

Systematics of real emissions: collinear limits

$$\left|\mathcal{M}^{(0)}_{a_1,a_2,\ldots}(p_1,\,p_2,\ldots)\right|^2 \simeq 4\pi\alpha_s\,\frac{2}{s_{12}}\left\langle\mathcal{M}^{(0)}_{a,\ldots}(p,\ldots)\right|\hat{\mathbf{P}}^{(0)}_{a_1a_2}(z,k_\perp;\epsilon)\left|\mathcal{M}^{(0)}_{a,\ldots}(p,\ldots)\right\rangle$$

The splitting functions depend on the flavours involved:



$$\hat{P}^{(0),\,\mu\nu}_{gg}(z,k_\perp;\epsilon) = 2C_A\left[-g^{\mu\nu}\left(\frac{z}{1-z}+\frac{1-z}{z}\right)-2(1-\epsilon)z(1-z)\frac{k_\perp^\mu k_\perp^\nu}{k_\perp^2}\right],$$

$$\hat{P}^{(0),\,\mu\nu}_{q\bar{q}}(z,k_\perp;\epsilon) = \hat{P}^{(0),\,\mu\nu}_{\bar{q}q}(z,k_\perp;\epsilon) = T_F\left[-g^{\mu\nu}+4z(1-z)\frac{k_\perp^\mu k_\perp^\nu}{k_\perp^2}\right],$$

$$\hat{P}^{(0),\,ss'}_{qg}(z,k_\perp;\epsilon) = \hat{P}^{(0),\,ss'}_{\bar{q}g}(z,k_\perp;\epsilon) = \delta^{ss'}\,C_F\left[\frac{1+z^2}{1-z}-\epsilon(1-z)\right],$$

$$\hat{P}^{(0),\,ss'}_{gq}(z,k_\perp;\epsilon) = \hat{P}^{(0),\,ss'}_{g\bar{q}}(z,k_\perp;\epsilon) = \hat{P}^{(0),\,ss'}_{qg}(1-z,k_\perp;\epsilon).$$

For e+e- example:
g collinear with quark

$$P_{qg}(z)$$

$$z = E_q/(E_q+E_g) = x_q/(x_q+x_g)$$

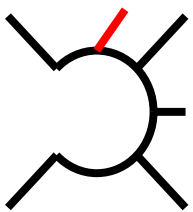
Next-to-leading order case

$$\hat{\sigma}^{(1)}_{ab} = \hat{\sigma}^{\mathrm{R}}_{ab} + \hat{\sigma}^{\mathrm{V}}_{ab} + \hat{\sigma}^{\mathrm{C}}_{ab}$$

KLN theorem
sum is finite for sufficiently inclusive observables and regularization scheme independent

Each term separately infrared (IR) divergent:

Real corrections:

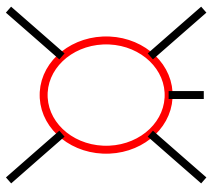


$$\hat{\sigma}^{\mathrm{R}}_{ab} = \frac{1}{2\hat{s}} \int \mathrm{d}\Phi_{n+1} \left\langle \mathcal{M}^{(0)}_{n+1} \middle| \mathcal{M}^{(0)}_{n+1} \right\rangle \mathrm{F}_{n+1}$$

Measurement function

Phase space integration over unresolved configurations

Virtual corrections:



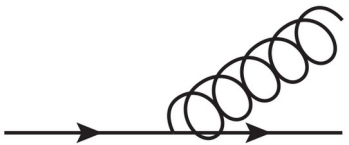
$$\hat{\sigma}^{\mathrm{V}}_{ab} = \frac{1}{2\hat{s}} \int \mathrm{d}\Phi_n \, 2\mathrm{Re} \left\langle \mathcal{M}^{(0)}_{n} \middle| \mathcal{M}^{(1)}_{n} \right\rangle \mathrm{F}_n$$

Integration over loop-momentum
(UV divergences cured by renormalization)

IR singularities in real radiation

$$\hat{\sigma}^{\mathrm{R}}_{ab} = \frac{1}{2\hat{s}} \int \mathrm{d}\Phi_{n+1} \left\langle \mathcal{M}^{(0)}_{n+1} \middle| \mathcal{M}^{(0)}_{n+1} \right\rangle \mathrm{F}_{n+1}$$

Finite function

$$\ni \mathrm{d}\Phi_1$$ 

$$\sim \int_0 \mathrm{d}E \mathrm{d}\theta \frac{1}{E(1-\cos\theta)} f(E, \cos(\theta))$$

Divergent

Regularization in Conventional Dimensional Regularization (CDR) $d = 4 - 2\epsilon$

$$\to \int_0 \mathrm{d}E \mathrm{d}\theta \frac{1}{E^{1-2\epsilon}(1-\cos\theta)^{1-\epsilon}} f(E, \cos(\theta)) \sim \frac{1}{\epsilon^2} + \ldots$$

Cancellation against similar divergences in

$$\hat{\sigma}^{\mathrm{V}}_{ab} = \frac{1}{2\hat{s}} \int \mathrm{d}\Phi_n \, 2\mathrm{Re} \left\langle \mathcal{M}^{(0)}_{n} \middle| \mathcal{M}^{(1)}_{n} \right\rangle \mathrm{F}_n$$

How to extract these poles? Slicing and Subtraction

Central idea: Divergences arise from infrared (IR, soft/collinear) limits → Factorization!

Slicing

$$\hat{\sigma}^{\mathrm{R}}_{ab} = \frac{1}{2\hat{s}} \int_{\delta(\Phi)\geq\delta_c} \mathrm{d}\Phi_{n+1} \left\langle \mathcal{M}^{(0)}_{n+1} \middle| \mathcal{M}^{(0)}_{n+1} \right\rangle \mathrm{F}_{n+1} + \frac{1}{2\hat{s}} \int_{\delta(\Phi)<\delta_c} \mathrm{d}\Phi_{n+1} \left\langle \mathcal{M}^{(0)}_{n+1} \middle| \mathcal{M}^{(0)}_{n+1} \right\rangle \mathrm{F}_{n+1}$$

$$\approx \frac{1}{2\hat{s}} \int_{\delta(\Phi)\geq\delta_c} \mathrm{d}\Phi_{n+1} \left\langle \mathcal{M}^{(0)}_{n+1} \middle| \mathcal{M}^{(0)}_{n+1} \right\rangle \mathrm{F}_{n+1} + \frac{1}{2\hat{s}} \int \mathrm{d}\Phi_n \, \tilde{M}(\delta_c) \mathrm{F}_n + \mathcal{O}(\delta_c)$$

$$\ldots + \hat{\sigma}^{\mathrm{V}}_{ab} = \text{finite}$$

Subtraction

$$\hat{\sigma}^{\mathrm{R}}_{ab} = \frac{1}{2\hat{s}} \int \left(\mathrm{d}\Phi_{n+1} \left\langle \mathcal{M}^{(0)}_{n+1} \middle| \mathcal{M}^{(0)}_{n+1} \right\rangle \mathrm{F}_{n+1} - \mathrm{d}\tilde{\Phi}_{n+1} \, \mathcal{S}\mathrm{F}_n \right) + \frac{1}{2\hat{s}} \int \mathrm{d}\tilde{\Phi}_{n+1} \, \mathcal{S}\mathrm{F}_n$$

$$\frac{1}{2\hat{s}} \int \mathrm{d}\tilde{\Phi}_{n+1} \, \mathcal{S}\mathrm{F}_n = \frac{1}{2\hat{s}} \int \mathrm{d}\Phi_n \, \mathrm{d}\Phi_1 \, \mathcal{S}\mathrm{F}_n$$

Phase space factorization
→ momentum mappings

Most popular
NLO QCD schemes:
CS [hep-ph/9605323],
FKS [hep-ph/9512328]

→ Basis of modern event simulation

Slicing and Subtraction

Slicing

- Conceptually simple
- Recycling of lower computations
- Non-local cancellations/power-corrections
  → computationally expensive
- Comparatively easy to extend to N3LO

Subtraction

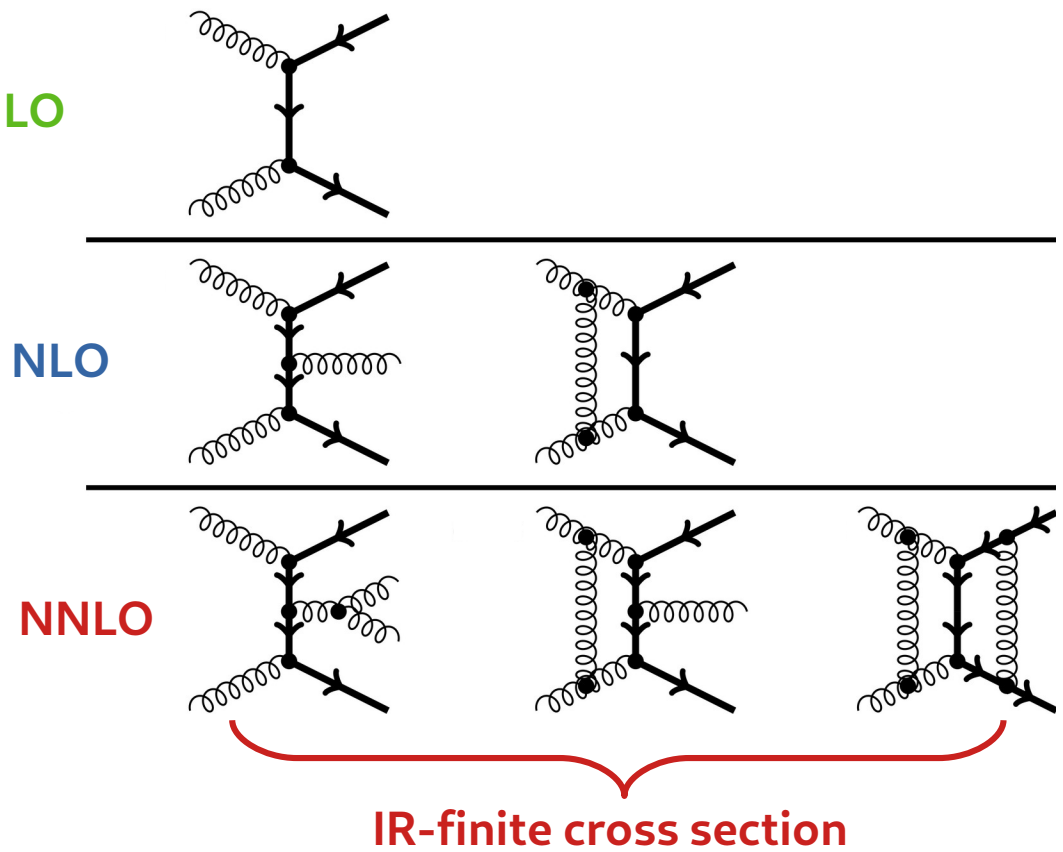
- Conceptually more difficult
- Local subtraction → efficient
- Better numerical stability
- Choices:
  - Momentum mapping
  - Subtraction terms
  - Numerics vs. analytic

NNLO QCD schemes

qT-slicing [Catani'07],
N-jettiness slicing [Gaunt'15/Boughezal'15]

Antenna [Gehrmann'05-'08],
Colorful [DelDuca'05-'15],
Sector-improved residue subtraction [Czakon'10-'14'19]
Projection [Cacciari'15],
Nested collinear [Caola'17],
Geometric [Herzog'18],
Unsubtraction [Aguilera-Verdugo'19],
...

NNLO QCD challenges: subtraction



How to achieve **infrared (IR) finite differential** cross sections at NNLO QCD?

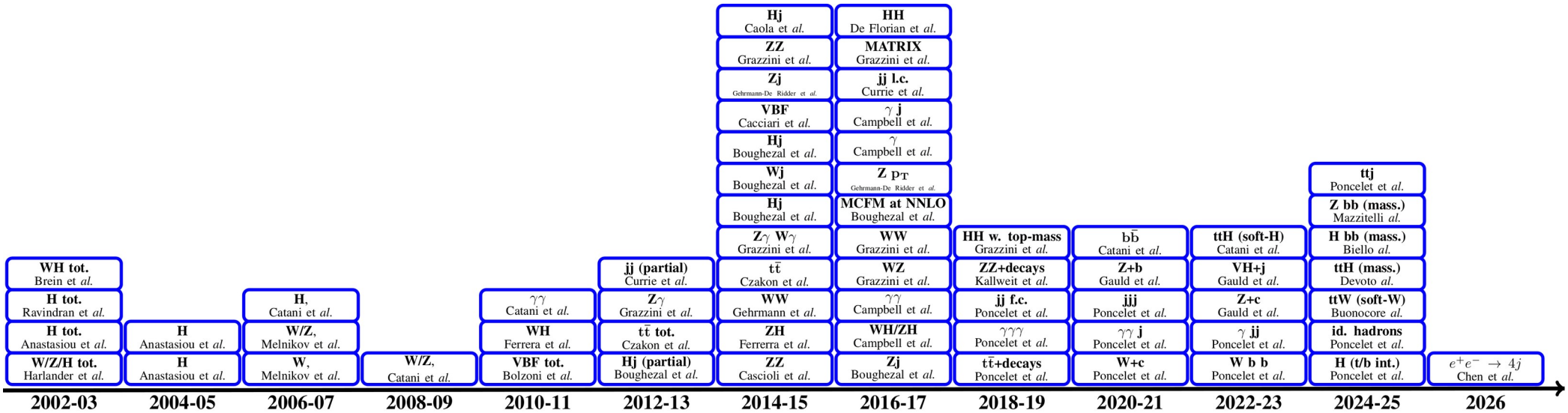
~20 years to solve this problem

qT-slicing [Catani'07], N-jettiness slicing [Gaunt'15/Boughezal'15], Antenna [Gehrmann'05-'08], Colorful [DelDuca'05-'15], Projetction [Cacciari'15], Geometric [Herzog'18], Unsubtraction [Aguilera-Verdugo'19], Nested collinear [Caola'17], Local Analytic [Magnea'18] Sector-improved residue subtraction [Czakon'10-'14,'19]

Automation of NNLO QCD subtraction:

- qT-slicing (massive/color singlet) in **MATRIX** & **MCFM** (public)
- Antenna subtraction in **NNLOJet** (public)
- Sector-improved residue subtraction in **Stripper** (private impl.)
- color singlets with nested-collinear in **history** (public)

Status of NNLO QCD cross section computations

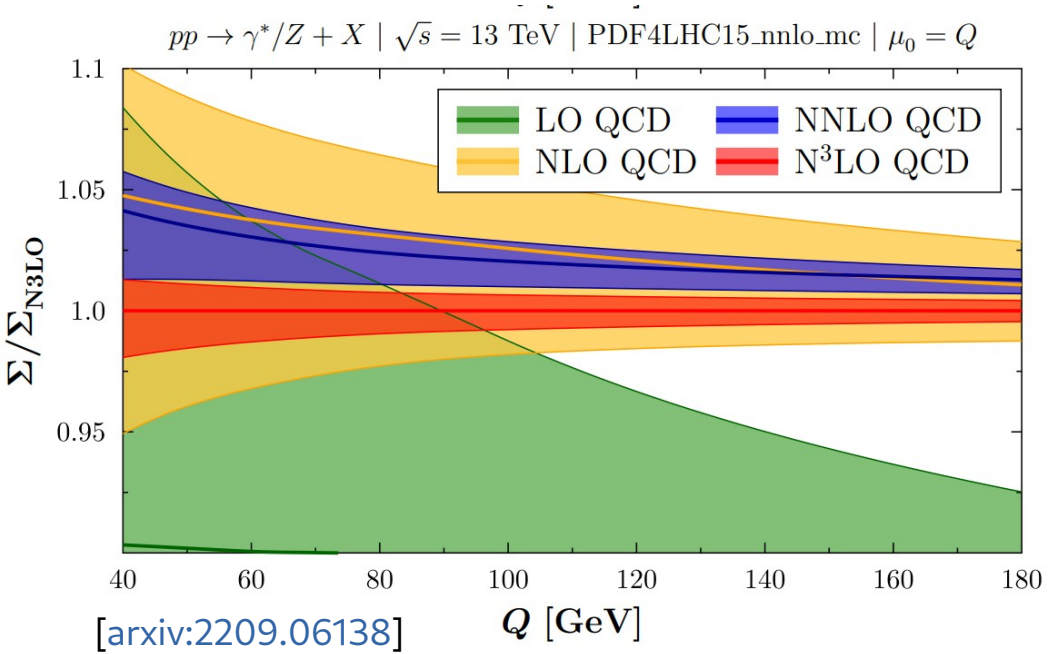


All 2 → 2 SM processes, 2 → 3+ only **limited by**

→ the available two-loop matrix elements

→ numerical/computational challenges

Drell-Yan higher order QCD cross sections



[arxiv:2209.06138]

Perturbative convergence:

- O(10%) correction at NLO
- O(1%) correction at NNLO
- O(1%) correction at N3LO? (PDF consistency?)

Being evermore precise is good, but

how to derive uncertainties?

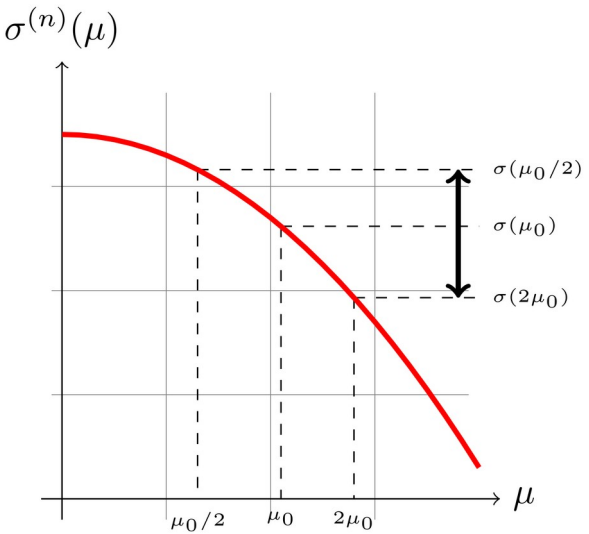
Theory uncertainties from scale variations

$$\mathrm{d}\sigma = \mathrm{d}\sigma^{(0)} + \alpha_s \mathrm{d}\sigma^{(1)} + \alpha_s^2 \mathrm{d}\sigma^{(2)} + \ldots \xrightarrow{\text{RGE}} \mu \frac{\mathrm{d}\sigma^{(n)}}{\mathrm{d}\mu} = \mathcal{O}\left(\alpha_s^{(n_0+n+1)}\right)$$

Of same order as the next dominant term → exploiting this to estimate size of $\mathrm{d}\sigma^{(n+1)}$

Scale variation prescription (ad-hoc and heuristic choice)

- choose 'sensible' μ_0
  - → principle of fasted apparent convergence: $\sigma^{(n)}(\mu_{\mathrm{FAC}}) = 0$
  - → principle of minimal sensitivity: $\mu \frac{\mathrm{d}\sigma(\mu)}{\mathrm{d}\mu}\Big|_{\mu=\mu_{\mathrm{PMC}}} = 0$
  - → ...
- vary with a factor (typically 2)
- take envelope as uncertainty



Scale variation approach

Change of scale = change of renormalisation scheme: $\tilde{\alpha}(\alpha) = \alpha(1 + b_0\alpha + b_1\alpha^2 + b_2\alpha^3 + \dots)$

$f(\alpha) = f_0 + f_1\alpha + f_2\alpha^2 + f_3\alpha^3 + \dots$ For QCD: $\alpha = \alpha_s(\mu_0)$ $\tilde{\alpha} = \alpha_s(\mu)$

$$b_0 = \frac{\beta_0}{2\pi}L \qquad b_1 = \frac{\beta_0^2}{4\pi^2}L^2 + \frac{\beta_1}{8\pi^2}L \qquad b_2 = \frac{\beta_0^3}{8\pi^3}L^3 + \frac{5\beta_0\beta_1}{32\pi^2}L^2 + \frac{\beta_2}{32\pi^3}L \qquad L = \ln\frac{\mu_0}{\mu}$$

$$\tilde{f}^{\mathrm{LO}}(\tilde{\alpha}) = \tilde{f}_0 = f_0$$

$$\tilde{f}^{\mathrm{NLO}}(\tilde{\alpha}) = \tilde{f}_0 + \tilde{f}_1\tilde{\alpha} = f_0 + \alpha f_1 + \boxed{\alpha^2 b_0 f_1} + \mathcal{O}(\alpha^3)$$

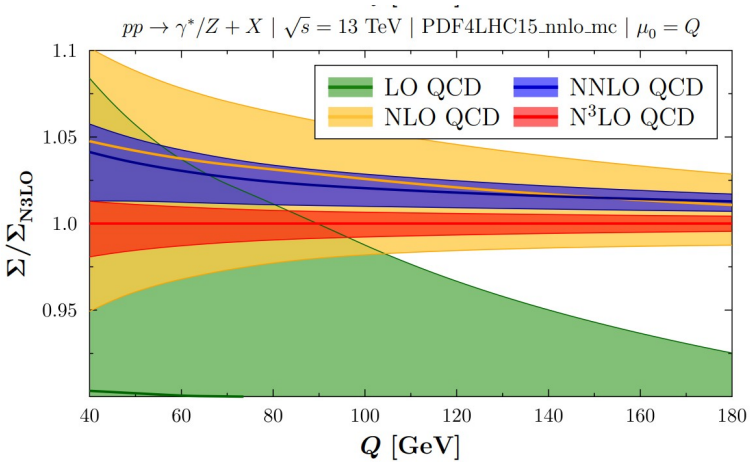
$$\tilde{f}^{\mathrm{NNLO}}(\tilde{\alpha}) = \tilde{f}_0 + \tilde{f}_1\tilde{\alpha} + \tilde{f}_2\tilde{\alpha}^2 = f_0 + \alpha f_1 + \alpha^2 f_2 + \boxed{\alpha^3(2b_0(f_2 - b_0 f_1) + b_1 f_1)} + \mathcal{O}(\alpha^4)$$

Scale variations as uncertainties can work …

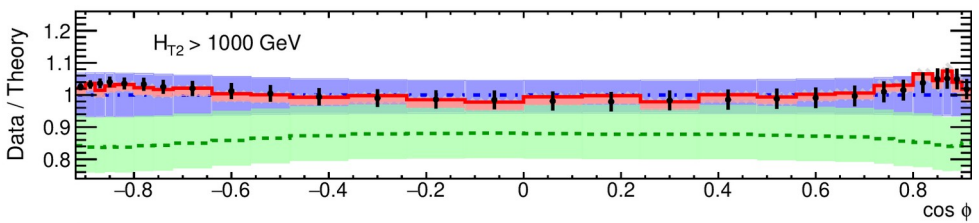
$$\sigma_{h_1 h_2 \to X} = \sum_{ij} \int_0^1 \int_0^1 \mathrm{d}x_1 \mathrm{d}x_2 f_{i,h_1}(x_1, \mu_F^2) f_{j,h_2}(x_2, \mu_F^2) \hat{\sigma}_{ij \to X}(\alpha_s(\mu_R^2), \mu_R^2, \mu_F^2) + \mathcal{O}\left(\frac{\Lambda^2}{Q^2}\right)$$

→ two scales: renormalisation and factorisation scale
→ conventional 7-point variations by a factor of 2

"Agreement within the variation envelope"

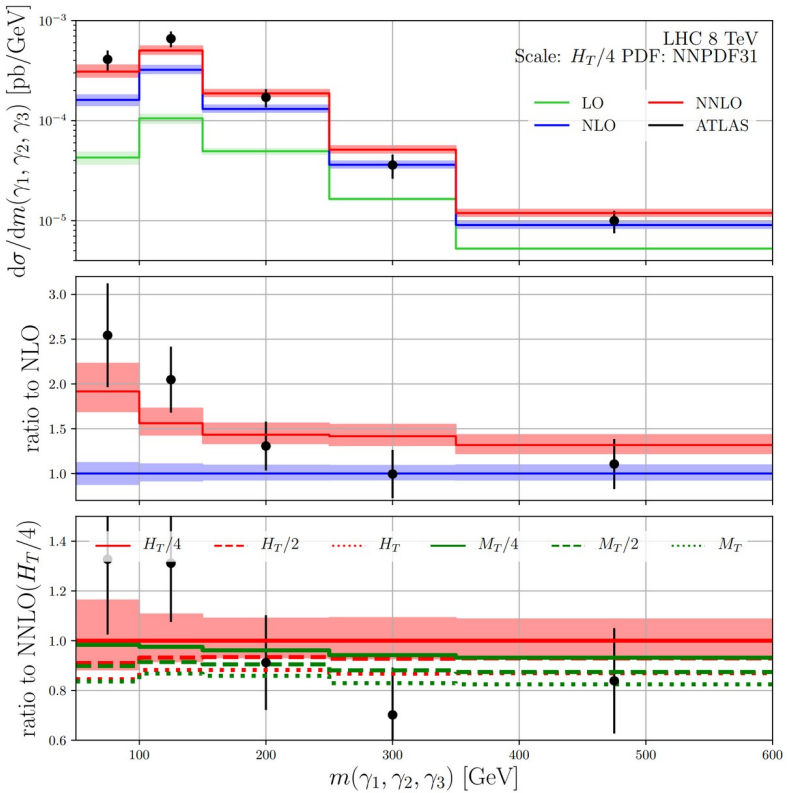


multi-jet cross section (TEEC)

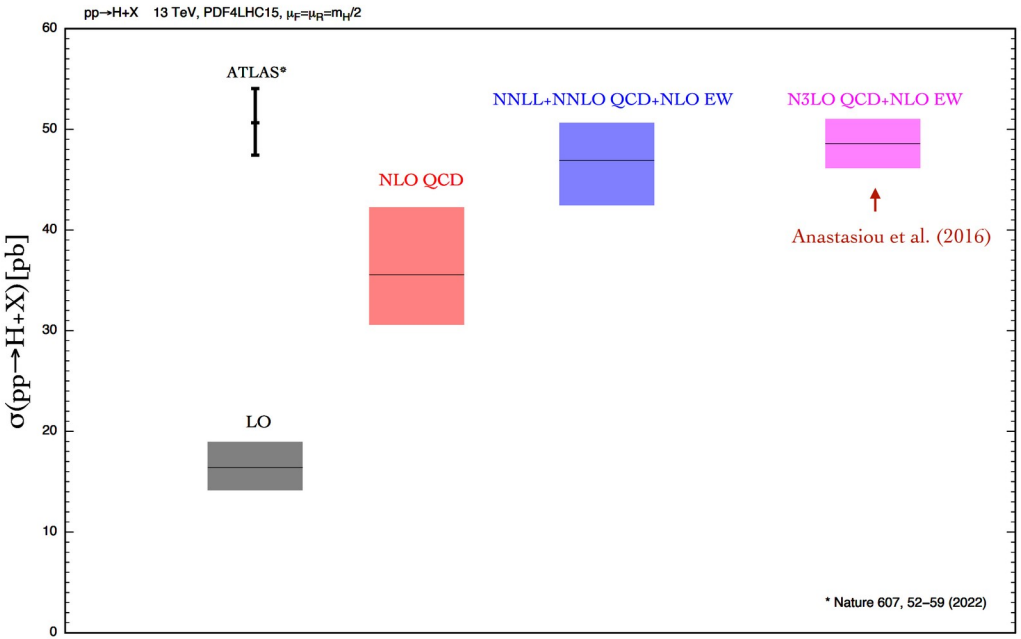


...sometimes :/

Three photon production



Higgs production



[talk by Grazzini]

NNLO QCD needed before “convergence” kicks in...

Short comings of scale variations

- not always reliable … however in most cases issues are understood/expected:
  new channels, phase space constraints, etc. → often we can design workarounds
- however, some issues are more fundamental:
  - → how to choose the central scale? → not a physical parameter, no 'true' value
    (Principle of fasted apparent convergence, principle of minimal sensitivity,…)
  - → how to propagate the estimated uncertainty, no statistical interpretation!
  - → what about correlations? Based on 'fixed form' of the lower orders and RGE.
- Alternatives are an active research topic:
  - Bayesian methods
  - Theory Nuisance Parameters
- with increasing precision this becomes more relevant…

Parton-showers and matching

Parton-showers → see Andrzej Siodmok lecture

Emission probability of in the collinear regime:

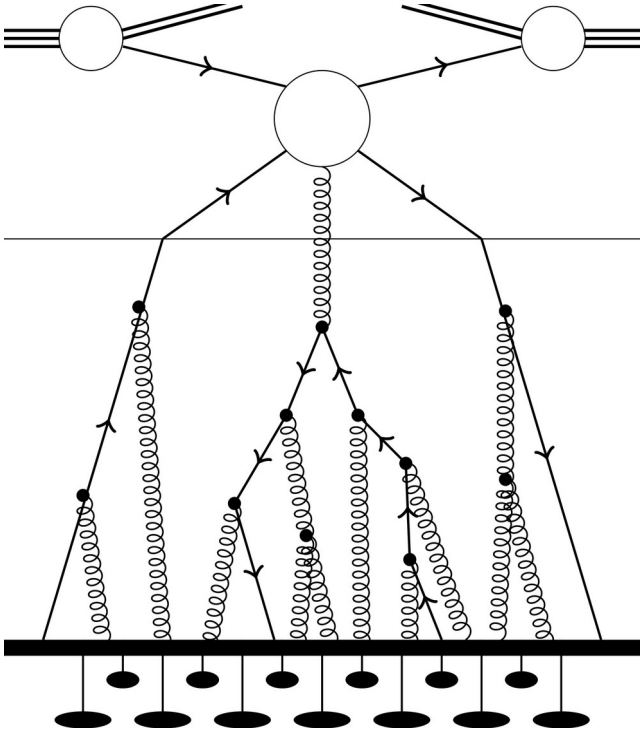
$$\left|\mathcal{M}^{(0)}_{a_1,a_2,\ldots}(p_1,p_2,\ldots)\right|^2 \simeq 4\pi\alpha_s \frac{2}{s_{12}} \left\langle \mathcal{M}^{(0)}_{a,\ldots}(p,\ldots)\middle|\hat{\mathbf{P}}^{(0)}_{a_1a_2}(z,k_\perp;\epsilon)\middle|\mathcal{M}^{(0)}_{a,\ldots}(p,\ldots)\right\rangle$$

Iteration with the Sudakov algorithm:

$$\sigma^{\text{incl}} \left(\Delta(t_0,\mu_Q^2) + \int_{t_0}^{Q} \frac{\mathrm{d}t}{t} \int \mathrm{d}z \frac{\alpha_s}{2\pi} P(z) \Delta(t,\mu_Q^2) + \ldots \right)$$

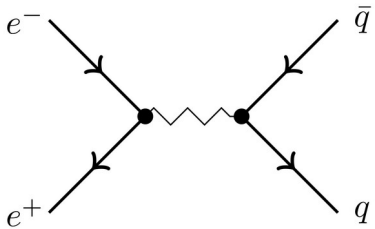
Sudakov form factor:

$$\Delta(t_0,t_1) = \exp\left(- \int_{t_1}^{t_0} \frac{\mathrm{d}t}{t} \frac{\alpha_s}{2\pi} \int_{z_{\min}}^{z_{\max}} \mathrm{d}z P(z) \right)$$

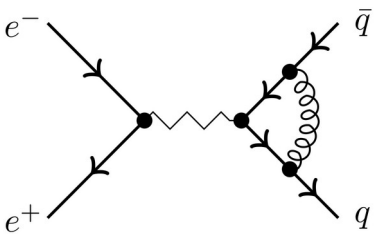


Matching at NLO

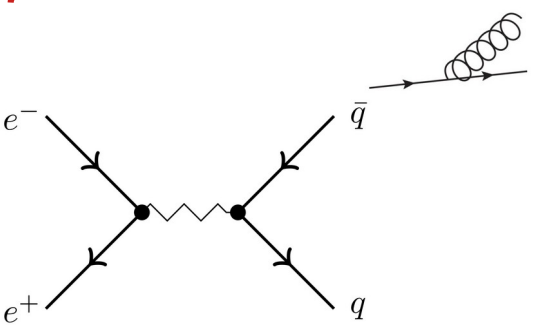
Expansion at α_s^0

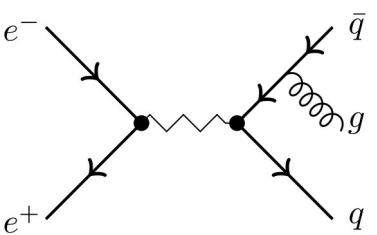


Expansion at α_s^1



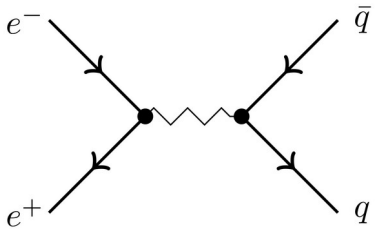
Double counting!



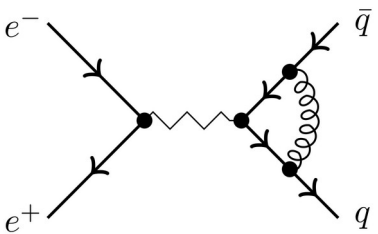


Matching at NLO

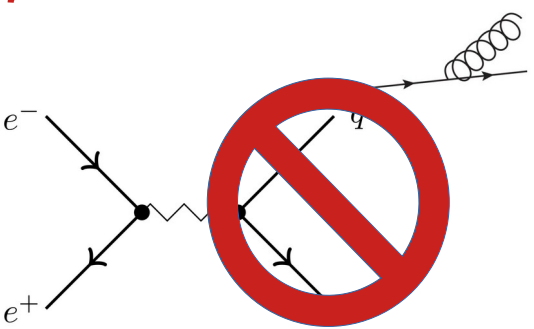
Expansion at α_s^0

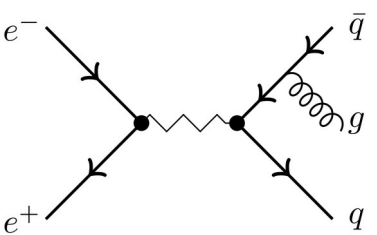


Expansion at α_s^1



Double counting!



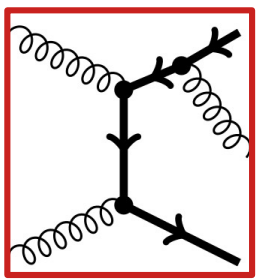


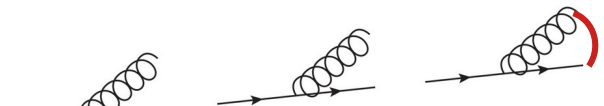
Beyond Leading Order: The Need for Matching

The challenge

Combine fixed-order with parton shower evolution
while **preserving** the precision/accuracy of both!

"Hard" scattering scale Q





v ← shower scale

parton shower approximation: $\alpha_S^i(v)\ln^j(v)$
but to all orders!

accuracy: LL, NLL, NNLL, …

Fixed-order: expansion in $\alpha_S(Q)$

→ cross section ~ $e^{\sum_{ij}\alpha_S^i(v)\ln^j(v)} \times$ (LO + NLO + NNLO + …)

A matching scheme needs to avoid double counting
of the logarithmic contributions!

Beyond Leading Order: The Need for Matching

NLO + Parton Shower Matching

- Standard tools: **POWHEG** and **MC@NLO**
- Implemented in all major frameworks: Herwig, SHERPA, Madgraph, Powheg-box,...
- Standard for most LHC analyses today.

Current state-of-the-art: NNLO+PS

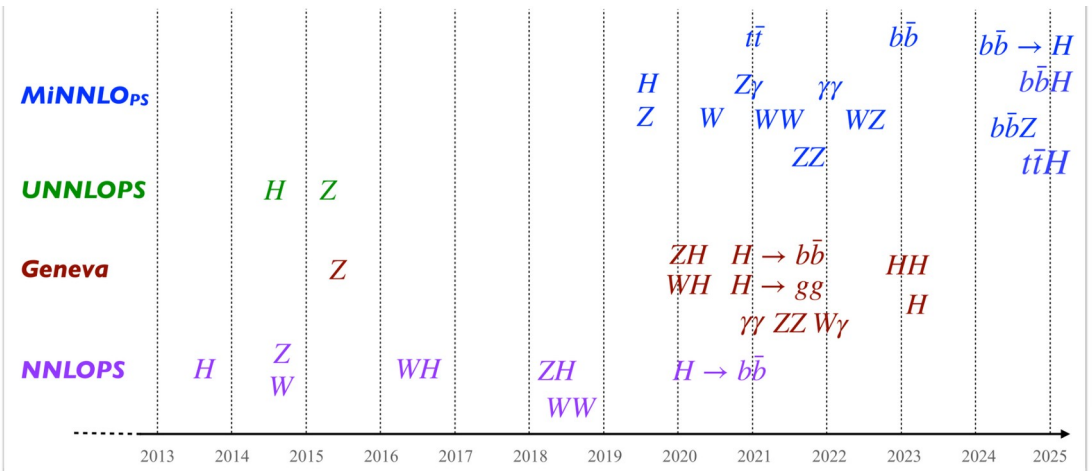
target to preserve LL accuracy so far

→ MiNNLOPS [Hamilton, Nason, Re, Zanderighi'13] [Monni, Nason, Re, Wiesemann, Zanderighi'20],

→ Geneva [Alioli, Bauer, Berggren, Hornig, Tackmann, Vermilion, Walsh, Zuberi'13-'25]

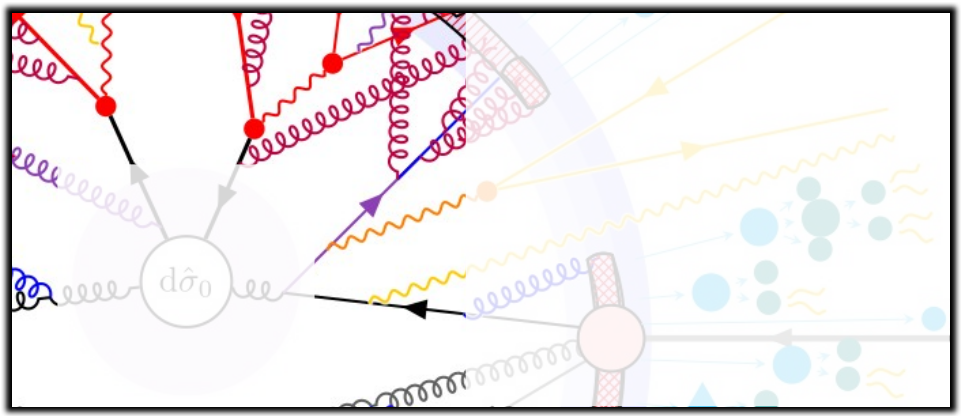
→ UN2LOPS [Hoche, Prestel'14]

→ Vincia [El-Menoufi,Preuss,Scyboz,Skands'24]



[talk by Wiesemann'25]

Higher accuracy parton showers



Steps towards better controlled accuracy in the parton shower

→ **NLL showers**

Not covered here:

→ Higher-order splitting kernels

→ Sub-leading color effects

→ Treatment of spin-correlation

→ Electroweak effects

PanScales [van Beekveld, Dasgupta, Dreyer, Ferrario Ravasio, Hamilton, Karlberg, Medves, Monni, Salam, Scyboz, Soto Ontoso, Soyez,Verheyen'18-'25]

Alaric [Assi,Herren, Höche, Krauss, Reichelt, Schönherr'22-'24]

Apollo [Preuss'24]

Deductor [Nagy, Soper'20]

Herwig [Forshaw, Holguin, Plätzer Bewick, Ferrario Ravasio, Richardson, Seymour'19-'21]

Jet substructure: why logarithmic accuracy matters

Jet sub-structure

- Example: Lund plane multiplicity by ATLAS
- comparison of different showers and analytic resummed predictions
- Data is sensitive to the shower model
  → how to **estimate shower uncertainties**?
  → why are the shower so different from the analytic resummation?

Outlook

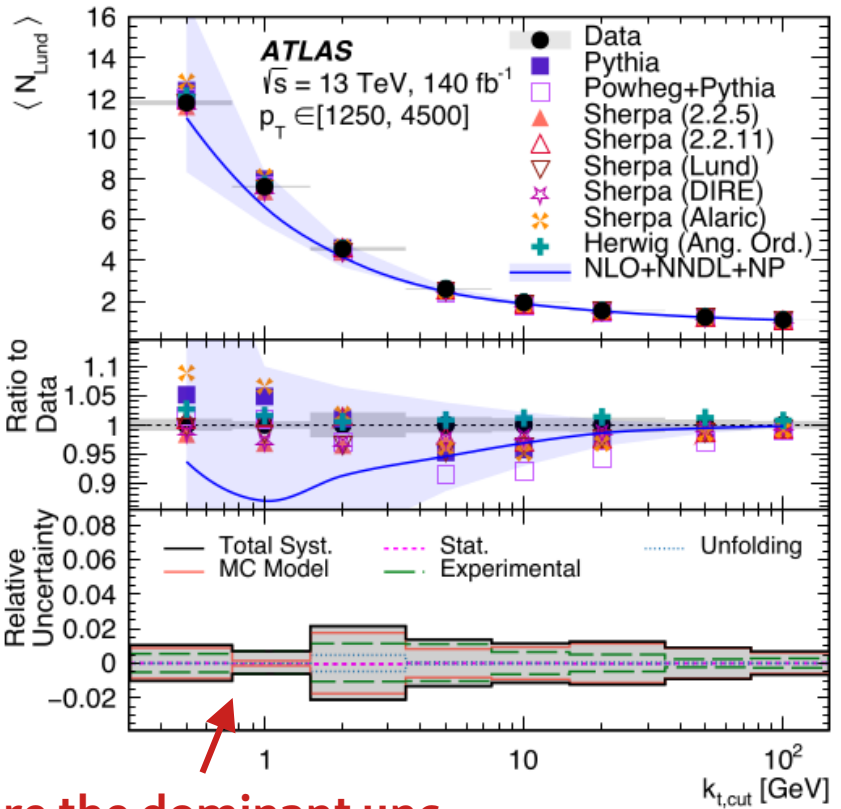
→ Implications for jet substructure measurements

→ everything that uses jet substructure techniques

→ flavour tagging

→ quark/gluon jet discrimination

→ jet calibration



MCs are the dominant unc.

Towards NNLL showers

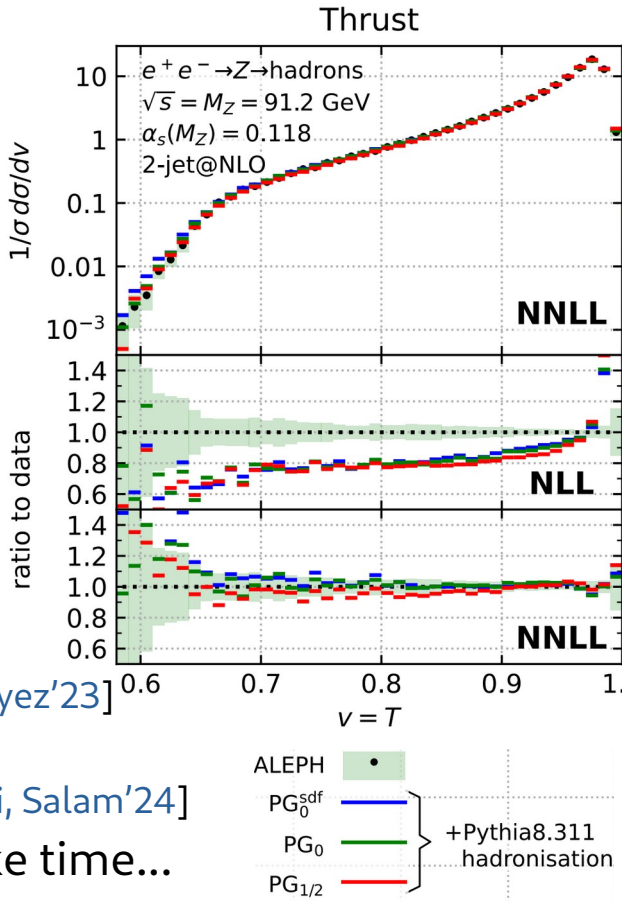
- For some e+e- event shapes already feasible

  → PanScales showers + Pythia

  → understood with $\alpha_S = 0.118$ at NLL → NNLL spot on

  → other showers might behave differently

- Take-home message:
  → high log accuracy can resolves modelling issues
  → increased consistency



Step-by-step towards NNLL showers come up

- Double soft-corrections [Ferrario Ravasio, Hamliton, Karlberg, Salam, Scyboz, Soyez'23]
- Soft-collinear emissions [PanScales'24]
- Triple-collinear emissions [van Beekveld, Dasgupta, el-Menoufi, Helliwell, Monni, Salam'24]

→ so far limited to special cases in e+e-, transfer to pp/DIS will take time…

Jets

Infrared safety of the measurement function

$$\sigma \sim \int \mathrm{d}\Phi_n |\mathcal{M}_n|^2 \to \int \mathrm{d}\Phi_n |\mathcal{M}_n|^2 \mathcal{F}_n(p_1, \ldots, p_n)$$

'Measurement function' defines
→ observables → cuts → jets
→ histograms → ...

KLN theorem: average over sufficient unresolved degrees of freedom
→ IR safe observables: well behaved* in the soft and collinear regions!

*the precise notion what properties are sufficient or equivalent is sometimes still a matter of debate

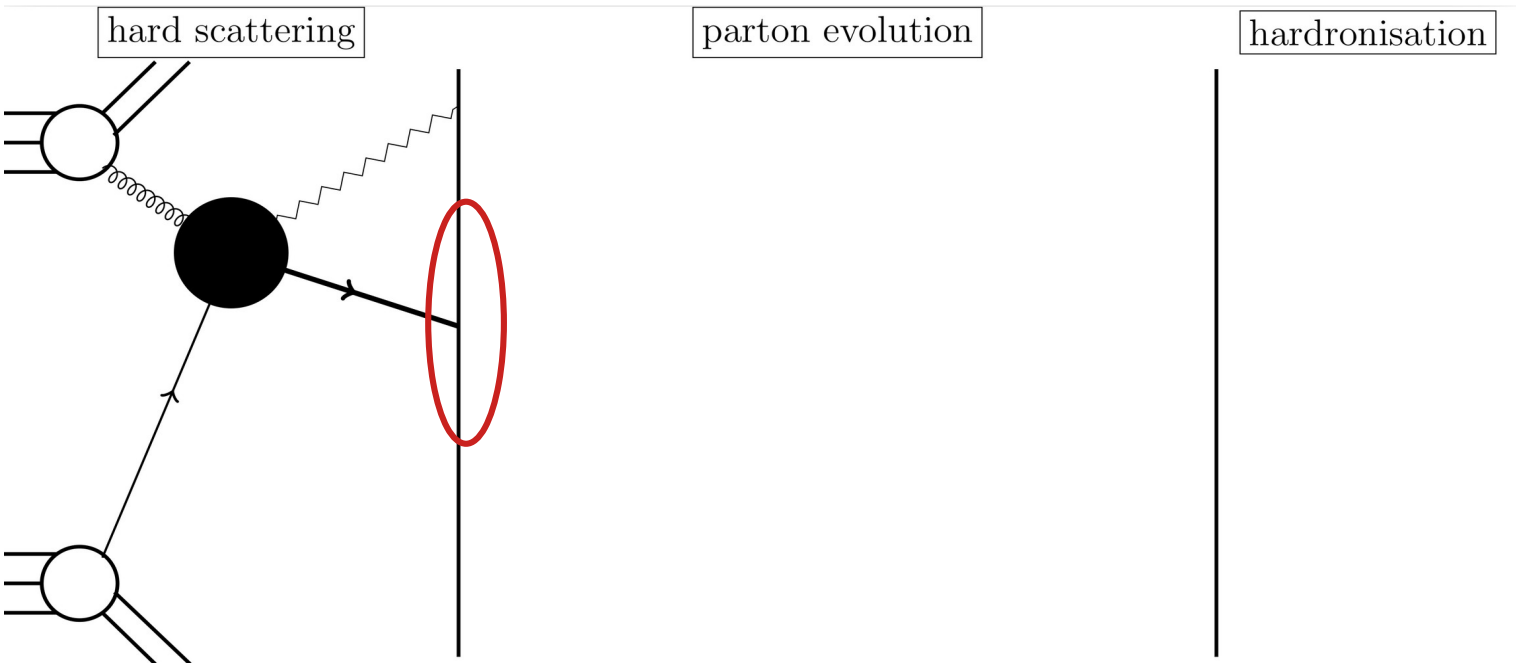
Soft limits:

$$\mathcal{F}_{n+1}(p_1, \ldots, p_{n+1}) \to \mathcal{F}_n(p_1, \ldots, \not{p}_i, \ldots, p_{n+1})$$

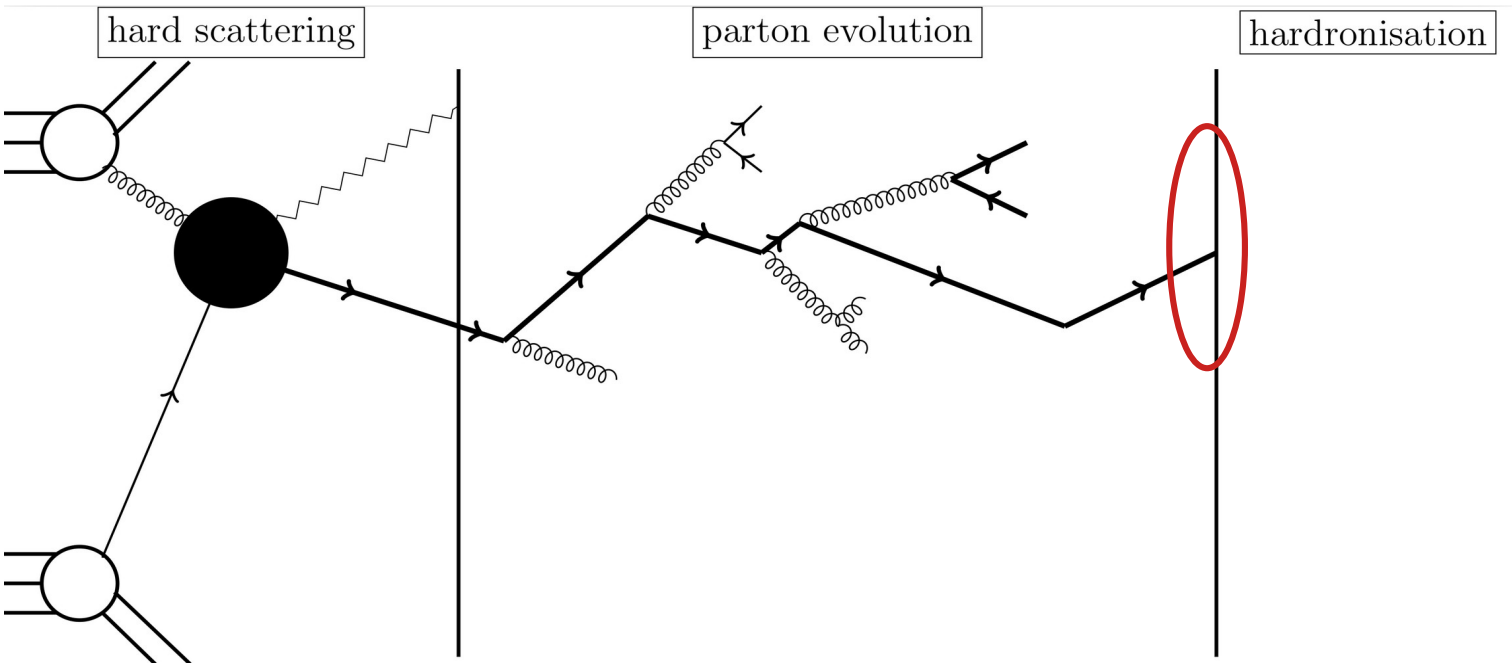
Collinear limits:

$$\mathcal{F}_{n+1}(p_1, \ldots, p_{n+1}) \to \mathcal{F}_n(p_1, \ldots, \not{p}_i, \ldots, \not{p}_j, \ldots, p_{n+1}, p_i + p_j)$$

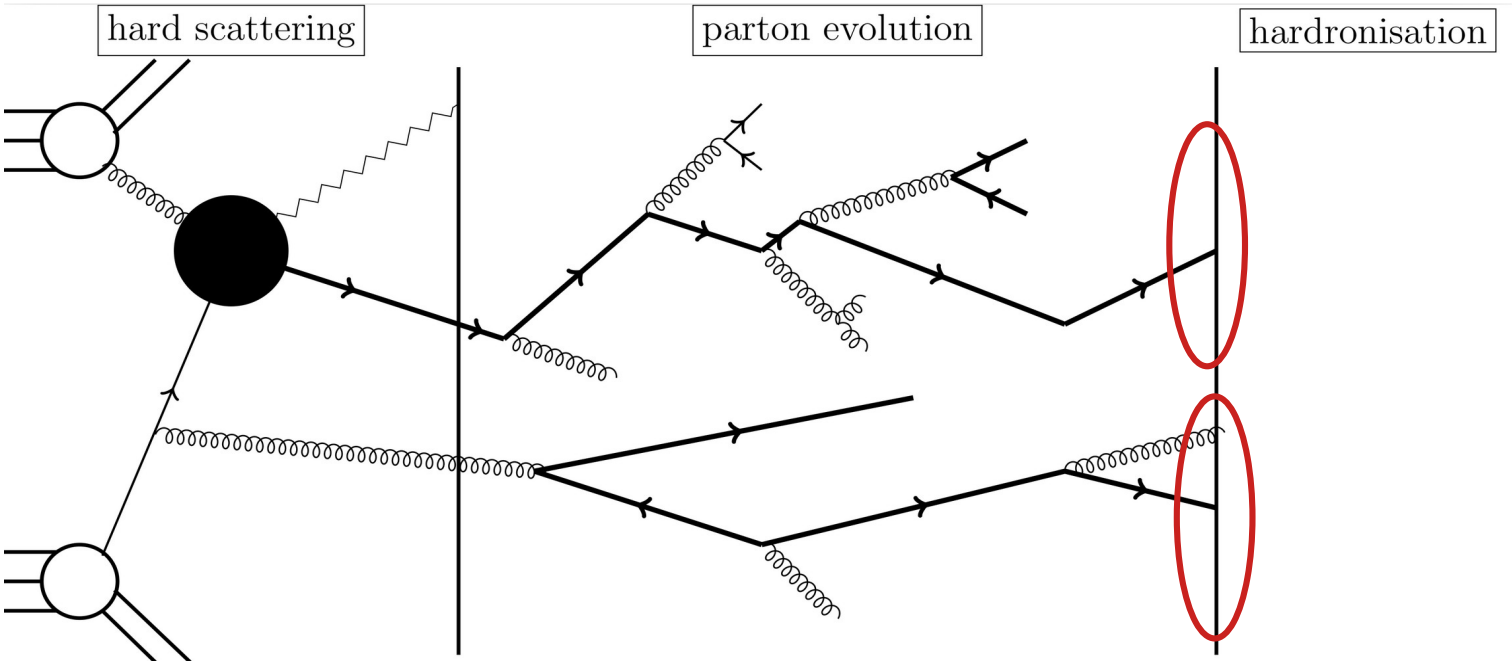
Jet formation



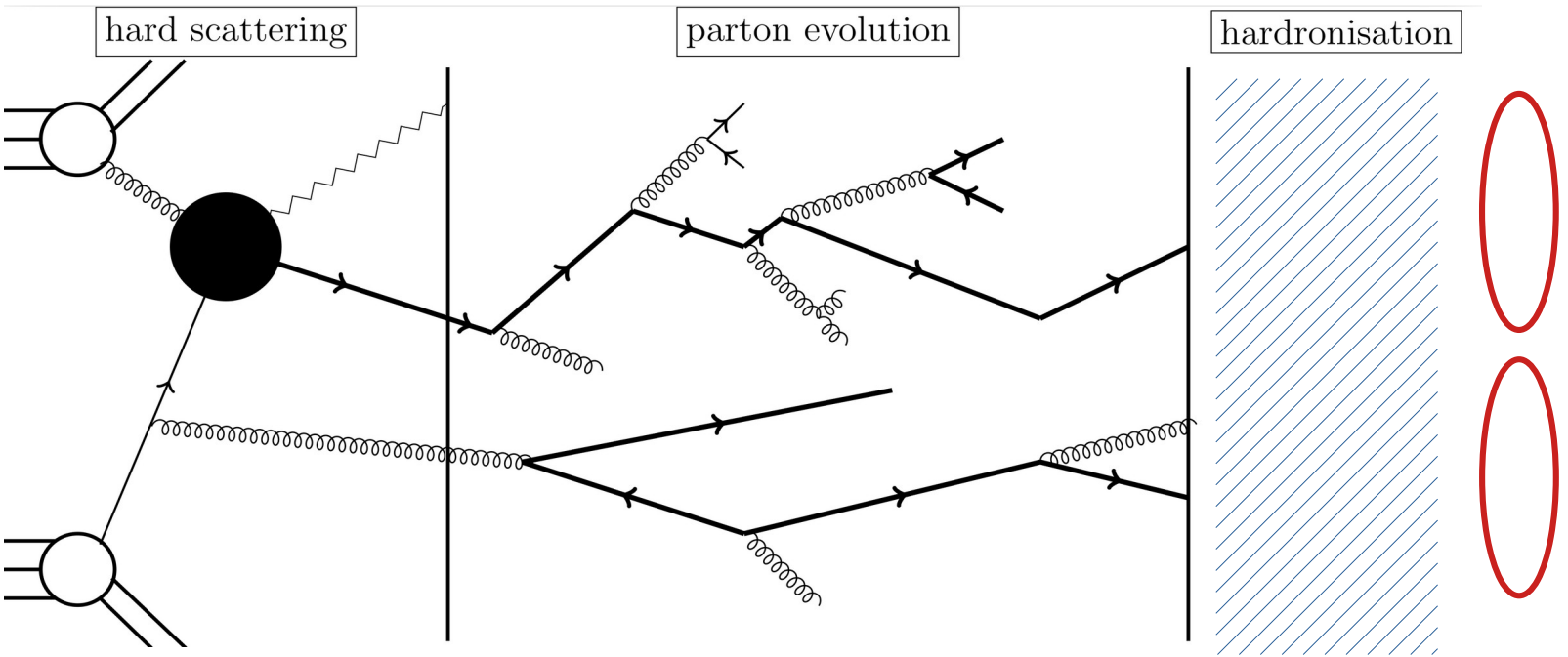
Jet formation



Jet formation



Jet formation



A look back: Snowmass accord 1990 …

Toward a Standardization of Jet Definitions

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Nicholas Hadley
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Zurich, Switzerland

Davison Soper
University of Oregon
Eugene, Oregon 97403

December 1990

A sensible jet definition should be:

1) Simple to implement in experimental analysis
2) Simple to implement in theoretical calculations
3) Defined at any order of perturbation theory
4) Yields finite cross section at any order of perturbation theory
5) Yields a cross section that is relatively insensitive to hadronization

Purpose: “undo” parton evolution to define the “hard scattering” process

A look back: Snowmass accord 1990 …

A sensible jet definition should be:

1) Simple to implement in experimental analysis
2) Simple to implement in theoretical calculations
3) Defined at any order of perturbation theory
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5) Yields a cross section that is relatively insensitive to hadronisation

For theory:

- Infrared collinear (IRC) safety
- Small sensitivity of 'inclusive observables' to parton-shower & hadronisation

Purpose: "undo" parton evolution to define the "hard scattering" process

Jets at the LHC

Many proposals of jet algorithms since ‘90:

- Cone-based algorithms: PxCone, midpoint, seedless, SISCone, …
- 2-to-1 recombination algorithms: C/A, Jade, kT, anti-kT, …

The **standard** algorithm for the LHC is

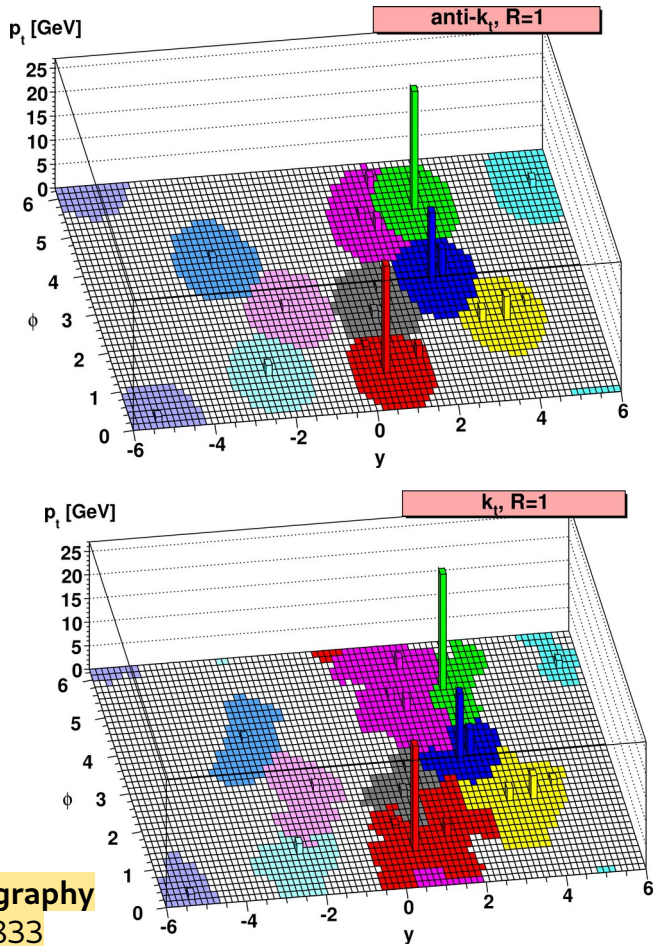
the **anti-kT algorithm**:

$$d_{ij} = \min(k_{T,i}^{-2}, k_{T,j}^{-2}) R_{ij}^2 \qquad d_i = k_{T,i}^{-2}$$

→ nice geometric properties

→ theoretically okay

→ insensitive to soft physics, pile up, etc.

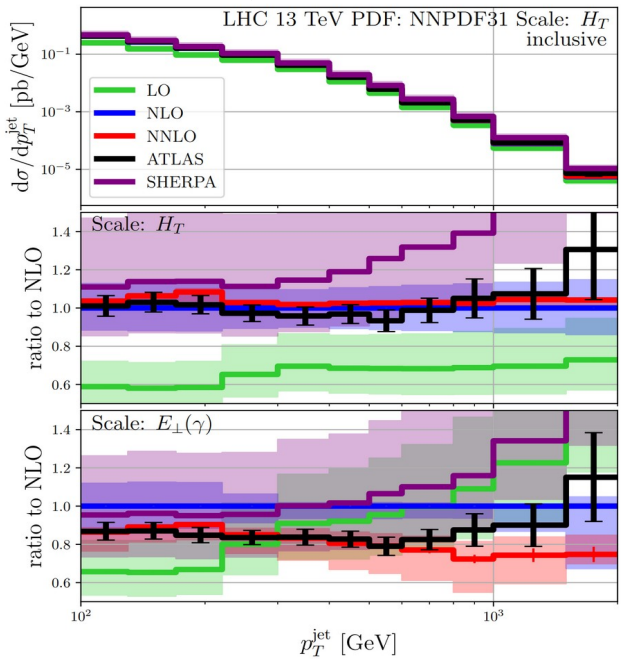


Towards Jetography
Salam 0906.1833

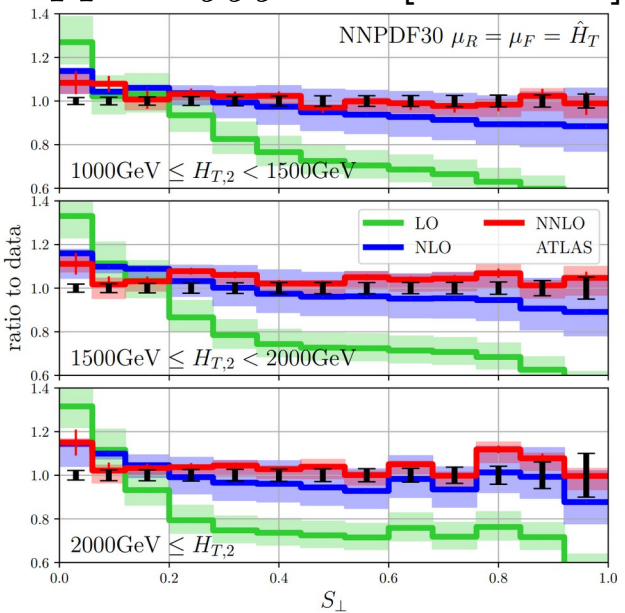
Precision comparisons of jet cross sections

Following these guidelines means that we can compare theory and experiment even though theorist talk about quarks+gluons and experimentalists about particles

$pp \to \gamma jj$ [2304.06682]



$pp \to jjj$ [2301.01086]



End of lecture

I hope you learn something about:

- structure of QCD
- a rough understanding of the impact of radiative corrections
- basics of higher-order computations
  - Renormalization
  - Phasespace and loop-integrals
  - IR singularities and how to treat them
- A bit about QCD phenomenology
  - Uncertainties
  - Parton showers
  - Jets

